

AKHIL A R

SOFTWARE ENGINEER (AI/ML)

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SUMMARY

I'm an AI Application Engineer and Full Stack Architect focused on connecting stochastic LLMs with deterministic, production-ready systems. With a background in Physics and a Master's in Computer Applications, I apply engineering principles to AI, emphasizing reliability, mathematical foundations, and security. I design comprehensive RAG pipelines, autonomous agents, and multi-LLM ecosystems (including OpenAI, Gemini, Groq, and Claude), supported by strong vector databases to ensure factual, hallucination-free results.

I have developed enterprise-level RAG platforms (RQA), autonomous NL2SQL agents, and resilient AI-driven data pipelines (COMPA, REPA). My tech stack includes LangChain, LlamaIndex, FastAPI, Node.js, React, SQL/NoSQL, and secure API design. As a Certified Ethical Hacker (CEH), I create AI systems that are Secure by Design, merging rapid full-stack execution with data science-level validation to address real-world business challenges at scale.

WORK EXPERIENCE

June 2025 – Present

Software Engineer Trainee AI Full Stack Focus

Myblocks | Ernakulam, India

- **Generative AI Architectures:** Created a Multi-LLM Business Intelligence Copilot and an Autonomous NL2SQL Engine using LangChain, achieving a 95% accuracy in natural language querying of complex SQL datasets.
 - **Enterprise RAG NLP Systems:** Developed the RQA pipeline to reduce document analysis hallucinations and launched a Multi-Portal News Agent to summarize thousands of articles into concise market briefs.
 - **Resilient Data Pipelines:** Built high-availability scraping engines to monitor competitors and assess market sentiment, using anti-bot techniques and Sentiment Analysis for real-time feedback.
 - **Scalable Full Stack Modules:** Developed a Vendor Management System backend with secure document storage, Role-Based Access Control RBAC, and automated supplier performance tracking.
 - **Security Optimization:** Enhanced API security with JWT authentication and input validation, adhering to Certified Ethical Hacker CEH standards to prevent data leakage and Prompt Injection.
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EDUCATION

Master of Computer Application

- Graduated: June, 2024

Bachelor of Science in Physics and Computer Application

- Graduated: March, 2020

PROJECTS

NL2SQL | Autonomous Natural Language to SQL Agent

- **Tech Stack:** LangChain, Groq (Llama 3), FastAPI, React, MySQL
 - Developed a deterministic AI agent capable of converting natural language queries into complex, executable SQL statements with an impressive 95% accuracy.
 - Integrated LangChain's SQL Toolkit for schema-aware prompting, enabling the system to automatically understand table relationships and foreign keys.
- Established a self-correcting feedback loop where execution errors trigger automatic regeneration, ensuring high reliability for non-technical users.

RQA | Enterprise RAG Document Analysis System

- **Tech:** Python, LangChain, Vector DB (Qdrant), Claude, React
 - Designed a Retrieval-Augmented Generation (RAG) pipeline to analyze and validate complex research documents.
 - Implemented Semantic Chunking and Vector Search to retrieve precise context, significantly reducing LLM hallucinations compared to standard prompting.
 - Integrated FastAPI for asynchronous document processing, allowing real-time Q&A and compliance scoring on uploaded PDFs.

COMPA | Competitive Market Intelligence Platform | Multi-LLM Orchestrator

- **Tech:** OpenAI, Gemini, Groq, Node.js, React, Chart.js
 - Created a comprehensive decision-support system that consolidates unstructured vendor data into strategic visual reports.
 - Developed a Multi-LLM Orchestration layer that directs simple tasks to smaller models (Groq) while delegating complex reasoning to larger models (GPT-4), optimizing both cost and latency.
 - Automated the generation of PDF insights using Prompt Engineering techniques to extract structured JSON data from raw text.
 - Designed a normalized database schema to store historical performance data, facilitating trend analysis and growth rate visualization.

REPA | Reputation Intelligence & Sentiment Engine

- **Tech:** Node.js, Puppeteer, Natural NLP, Sentiment.js, MySQL
 - Engineered a robust scraping pipeline to monitor brand sentiment across Google Maps and review portals, incorporating automated CAPTCHA handling.
 - Integrated an NLP Sentiment Analysis workflow to classify feedback trends and automatically draft context-aware responses using GenAI.
 - Reduced manual reputation management efforts by 70% through real-time alerts and automated reporting.

Multi-Portal News Agent | Automated Market Intelligence

- **Tech:** Python, BeautifulSoup, LangChain, Summarization Chains
 - Developed an autonomous news aggregator that collects articles from various sources to identify market trends and key entities.
 - Utilized LangChain's Map-Reduce summarization chain to condense extensive content into concise executive briefs.

Implemented specific Entity Extraction (NER) pipelines to track mentions of VIPs and organizations across the news landscape.

SKILLS

- **Generative AI:** RAG, LLM Orchestration, LangChain, LlamaIndex, Agents, Prompt Engineering.
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- **Models & Tools:** OpenAI (GPT-4), Claude 3, Groq (LPU), Gemini, Ollama, Hugging Face.
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- **Backend Engineering:** Python (FastAPI), Node.js (Express), Microservices, REST APIs.
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- **Data & Database:** SQL (MySQL), NoSQL (MongoDB), Vector DB (Qdrant).
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- **Frontend:** React.js, TypeScript, Tailwind CSS, Data Visualization (Chart.js).
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- **DevOps & Security:** Docker, Git/GitHub, AWS, CEH (Ethical Hacking), OWASP Security.